

The Shioda pairing by Néron evaluation

1. Construction

Let X be a smooth proper connected curve over $\mathbb{C}$, with generic point $\eta : \operatorname{Spec} K \rightarrow X$. Let E/K be an elliptic curve with unit section $O : \operatorname{Spec} K \rightarrow E$, dual E^t , and identification $\alpha : E \xrightarrow{\sim} E^t$, $Q \mapsto [\mathcal{O}_E(Q - O)]$. Put

$$\mathcal{A} = \eta_*(E_{\mathbb{Q}}) \quad \text{in } \operatorname{Shv}_{\text{ét}}(\operatorname{Sm}/X; \operatorname{Mod}_{\mathbb{Q}}).$$

Let $\mathcal{N}, \mathcal{N}^t$ be the Néron models, $\mathcal{N}^0 \subset \mathcal{N}$ the fibrewise identity component, and $e_{X, \mathbb{Q}}$ rational Néron evaluation [Suz, Proposition 4.7]. Define

$$b_X : \mathcal{A}^{\otimes 2} \xrightarrow[\alpha]{\sim} (\mathcal{N}^0)_{\mathbb{Q}} \otimes \mathcal{N}_{\mathbb{Q}}^t \xrightarrow{e_{X, \mathbb{Q}}} \mathbb{G}_{m, X, \mathbb{Q}}[1]. \quad (1)$$

Let Γ_X denote étale global sections and put $V = E(K) \otimes \mathbb{Q} \simeq \Gamma_X \mathcal{A}$. Define

$$\mathcal{H}_E : V^{\otimes 2} \longrightarrow \Gamma_X(\mathcal{A}^{\otimes 2}) \xrightarrow{-\Gamma_X(b_X)} \Gamma_X(\mathbb{G}_{m, X, \mathbb{Q}}[1]) \xrightarrow{\sim} \operatorname{pic}(X)_{\mathbb{Q}}. \quad (2)$$

The first arrow uses lax monoidality of Γ_X ; the last is the Picard identification by line bundles and their isomorphisms. Write $\mathcal{H}_E(P, Q)$ for its value on $P \otimes Q$, and set $h_E(P, Q) = \deg \mathcal{H}_E(P, Q)$.

2. Proofs

Lemma 1. *There are canonical equivalences*

$$(\mathcal{N}^0)_{\mathbb{Q}} \xrightarrow{\sim} \mathcal{N}_{\mathbb{Q}} \xrightarrow{\sim} \mathcal{A} \xleftarrow{\sim} (\eta_* E)_{\mathbb{Q}}. \quad (3)$$

Proof. Let T be a connected smooth K -scheme, with generic point $j : \operatorname{Spec} F \rightarrow T$. Weil extension [Mil, I, Theorem 3.2] identifies sections of E on every connected étale $V \rightarrow T$ with their generic restrictions. Positive Galois cohomology of a discrete rational module vanishes. Hence, on the small étale site of T ,

$$E_{T, \mathbb{Q}} \simeq j_* E_{F, \mathbb{Q}}, \quad H_{\text{ét}}^q(T, E_{T, \mathbb{Q}}) = 0 \quad (q > 0).$$

Apply this to the components of U_K for $U \in \operatorname{Sm}/X$. The Néron mapping property gives $\mathcal{A} \simeq \mathcal{N}_{\mathbb{Q}}$. Commutation of étale cohomology with filtered colimits [Stacks, Lemma 59.51.4] gives $(\eta_* E)_{\mathbb{Q}} \simeq \mathcal{A}$. Finally, $\mathcal{N}/\mathcal{N}^0$ has finite stalks and is supported at finitely many closed points, so $(\mathcal{N}^0)_{\mathbb{Q}} \simeq \mathcal{N}_{\mathbb{Q}}$. $\square$

Lemma 2. *There are canonical equivalences*

$$\Gamma_X \mathcal{A} \simeq E(K) \otimes \mathbb{Q}, \quad \Gamma_X(\mathbb{G}_{m, X, \mathbb{Q}}[1]) \simeq \operatorname{pic}(X)_{\mathbb{Q}}.$$

Proof. The first is rational Galois acyclicity, as in Lemma 1. The Picard spectrum is the spectrum of line bundles and their isomorphisms under tensor product. Descent identifies it with the connective global sections of $\mathbb{G}_{m, X}[1]$ [Stacks, Theorem 59.24.1]. Since X is a smooth curve over $\mathbb{C}$, $H_{\text{ét}}^q(X, \mathbb{G}_m) = 0$ for $q \geq 2$ [Stacks, Theorem 59.68.5], so these global sections are already connective. Rationalization commutes with them by [Stacks, Lemma 59.51.4]. $\square$

Comparison with the surface definition

Let $f : S \rightarrow X$ be the relatively minimal regular elliptic surface with generic fibre E . Its smooth locus is $\mathcal{N}$ [BLR, §1.5, Proposition 1]. For $Q \in E(K)$, also denoting its section by Q , define the line bundle

$$L_Q = \mathcal{O}_S(Q - O) \otimes f^*(O^* \mathcal{O}_S(Q - O))^\vee. \quad (4)$$

It has its induced trivialization along O .

Let F be the fibre class, and let $T \subset \text{NS}(S)_\mathbb{Q}$ be generated by O, F and the components of singular fibres disjoint from O . The intersection form is nondegenerate on T . Shioda's homomorphism $\varphi : E(K) \otimes \mathbb{Q} \rightarrow T^\perp$ is the orthogonal projection of the section class, and his pairing is

$$\langle P, Q \rangle_{\text{Sh}} = -\varphi(P) \cdot \varphi(Q) \quad [\text{SS}, \S\S 11.3\text{--}11.8]. \quad (5)$$

Theorem 3. *For all $P, Q \in E(K)$,*

$$h_E(P, Q) = \langle P, Q \rangle_{\text{Sh}}.$$

If $P \in \mathcal{N}^0(X)$, there is a canonical equivalence

$$\mathcal{H}_E(P, Q) \simeq (P^* L_Q)^\vee \quad \text{in } \Omega^\infty \text{pic}(X)_\mathbb{Q}, \quad (6)$$

where the right side is the image of the indicated line bundle.

Proof. (1) The restriction of L_Q to $\mathcal{N}^0$ extends the generic line corresponding to $\alpha(Q)$, with its trivialization along O . Any two such extensions differ by a vertical divisor. Each fibre of $\mathcal{N}^0 \rightarrow X$ is reduced and irreducible, so this divisor is pulled back from X ; the trivialization along O gives a unique identification of the extensions. Thus Néron evaluation at $(P, \alpha(Q))$ is $P^* L_Q$. The minus sign in (2) gives (6), and therefore

$$h_E(P, Q) = -\deg P^* L_Q = -(P - O) \cdot (Q - O) \quad (P \in \mathcal{N}^0(X)). \quad (7)$$

(2) For $P \in \mathcal{N}^0(X)$, the divisor $P - O$ is orthogonal to every vertical divisor. Hence

$$\varphi(P) = P - O - ((P - O) \cdot O)F.$$

Since $\varphi(Q) - (Q - O)$ is vertical and $(Q - O) \cdot F = 0$, (5) and (7) give

$$\langle P, Q \rangle_{\text{Sh}} = -(P - O) \cdot (Q - O) = h_E(P, Q).$$

(3) Choose $n > 0$ annihilating $\mathcal{N}/\mathcal{N}^0$. Then $nP \in \mathcal{N}^0(X)$ for every $P \in E(K)$. Bilinearity gives

$$n h_E(P, Q) = h_E(nP, Q) = \langle nP, Q \rangle_{\text{Sh}} = n \langle P, Q \rangle_{\text{Sh}}.$$

Division by n proves the assertion. □

References

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